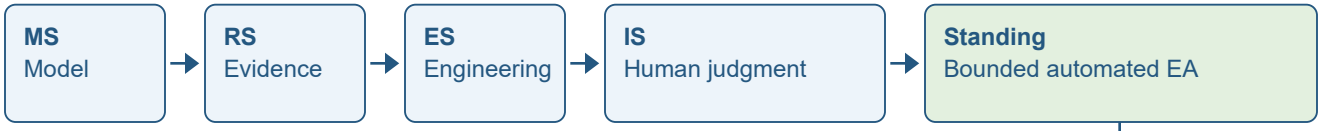
Governance of Autonomous Execution

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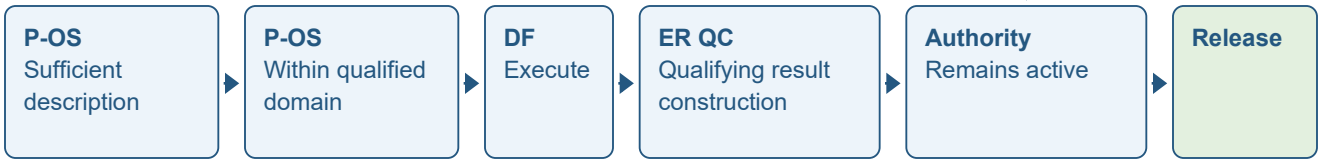
The governance objective is to establish when autonomous execution is justified, determine whether that justification continues to hold as conditions change, and withdraw authority when it no longer does. The architecture distinguishes three connected responsibilities: establishing Standing, governing individual decisions under that Standing, and improving the Decision Node through continuing engineering. Keeping these responsibilities distinct allows a problem to receive the response its scope warrants.

Establish and maintain Standing and autonomous Execution Authority



Sufficiency requirements and independent authorization are expanded below.

Process an individual candidate under Standing



Admission: both P-OS questions

Operating evidence + realized outcomes feed reassessment and continuing engineering

Candidate deficiency: HOLD. Governing ES failure or withdrawn IS: revoke Standing / withdraw EA.

The upper progression establishes bounded authority; the lower progression applies it to individual prediction candidates. The two P-OS questions form candidate admission. ER validity and result construction are distinct from the final check that authority remains active. These are related governance responsibilities, not a fresh MS-through-IS assessment for every candidate.

The Real Property Valuation Decision Node (RPVDN) provides a concrete setting for exploring this architecture. The same configured Decision Function serves two purposes. The LOOCV procedure uses the configured DF to produce held-out **sale-price estimates**. QC, calculation, and aggregation turn those outputs into metrics characterizing the evidence state for assessment against prospectively selected Engineering Sufficiency standards. Operational execution produces a **sale-price prediction** for an individual candidate using its eligible evidence state.

These outputs have distinct roles: a candidate's prior prediction is not carried into the LOOCV metrics of a later state. If the property subsequently enters that state, its held-out estimate is produced through fresh LOOCV executions. The laboratory examines empirical performance, execution integrity, and a change in state. The resulting governance framework extends beyond what those experiments demonstrate: it proposes how qualification, authorization, candidate controls, and feedback could operate together. It is a starting point for further engineering and research, with important controls still requiring validation.

Establish Standing

The first progression in the diagram is Model Sufficiency, Representation Sufficiency, Engineering Sufficiency, Interpretive Sufficiency, and Standing. Each addresses a different part of the justification for autonomous execution.

Model Sufficiency (MS) asks whether the bounded depiction of reality is adequate for the intended use. For RPVDN, this includes the selected property information, the eligible completed-sale population, and the rules by which that population changes. A Decision Function can operate consistently within a model that omits consequential aspects of reality. The adequacy of those boundaries therefore requires attention in its own right.

Representation Sufficiency (RS) asks whether the information presented to the Decision Function faithfully preserves and conveys the intended evidence identified by the Model. In RPVDN, the Decision Surface Snapshot (DSS) and structured quantitative table must represent the selected evidence without consequential errors, omissions, or misrepresentations. Faithful representation supports the engineering case; it does not establish that the model captures everything material to a valuation.

Engineering Sufficiency (ES) asks whether the affirmative engineering case supports the defined Decision Function within its qualified domain. Outcome-redacted Leave-One-Out Cross-Validation (LOOCV) supplies empirical evidence for this judgment in RPVDN. The observed errors, ensemble variability, execution controls, and limitations must be assessed against the governing requirements. A favorable average alone cannot establish sufficiency for every contemplated use.

Before Standing is established, this engineering assessment includes whether the selected methods and controls support the intended claims. Human review may lead to further experiments, revised requirements, and renewed ES assessment before IS permission is granted. Once automated Execution Authority is engaged, routine checks apply the prospectively established requirements of that approved engineering case. They do not repeat the entire human assessment or require a fresh IS judgment for each candidate. Existing IS permission must remain in force; material changes or consequential observations can reopen the broader engineering assessment and authorization judgment.

Interpretive Sufficiency (IS) is the authorized human steward's independent judgment that the evidence, limitations, and operating conditions justify the contemplated consequence. Affirmative ES does not compel permission. The steward may judge the remaining uncertainty unacceptable, even when the engineering requirements are satisfied. Conversely, IS cannot waive failure of a governing ES requirement.

Affirmative ES and IS permit **Standing**: bounded, continuing, revocable authorization carrying Execution Authority. Its scope must be identifiable, including the configuration, qualified domain, applicable evidence state, and conditions under which automated execution may continue. Standing makes an authorized automated path available; it does not establish that every individual candidate may enter or complete that path.

Govern Individual Predictions

Candidate admission: P-OS. The next progression begins with a candidate and Prospective Operating Sufficiency (P-OS). This gate asks whether the particular property is sufficiently described and falls within the qualified operating envelope. These are separate questions. Missing required evidence may be repairable; a fully described property may still fall outside the domain for which automated valuation is justified. The applicable requirements and boundaries must be specified before evaluating the candidate.

A P-OS failure can place that candidate on HOLD while evidence is obtained, corrected, or reviewed, or while an appropriate alternative path is considered. It need not change Standing for other eligible candidates. Nor should an operating boundary be relaxed ad hoc merely because the Decision Function can produce an answer outside it.

Held-out LOOCV subjects also require evidence-completeness checks, and their execution records undergo QC before admission to the qualifying ensemble. Membership in the characterization population follows the Model's eligibility rules. These checks do not constitute the candidate-specific P-OS operating-envelope assessment or place held-out estimates on the prospective prediction-release path shown here.

DF execution, ER QC, and result construction. Admission is followed by DF execution and Experimental Record (ER) quality control. In the laboratory, only qualifying ERs entered the ten-run ensemble; excluded executions and their replacements remained part of the inspectable record. Result-handling controls also need to check extraction, qualifying ensemble membership, aggregation, and the records presented for release. Candidate admission therefore does not guarantee a releasable result. An invalid ER may require exclusion and replacement. Failure to assemble an acceptable ensemble under the governing operational rules may require HOLD or escalation for that candidate.

This is a second place where the level of response matters. An isolated ER failure does not necessarily invalidate the qualified configuration or withdraw authority for unrelated candidates. Repeated failures, shared defects, or evidence of a broader control weakness can have wider implications. A deployment would need explicit rules for retries, release, escalation, and the circumstances requiring system review. The laboratory's QC practice does not itself validate a complete operational protocol.

Before release: continuing authority. A final governance check should confirm that Standing remains active, including the steward's continuing IS permission, and that no intervening change has invalidated the conditions supporting this candidate's execution. A completed calculation does not preserve authority that has since been withdrawn. The governing procedure must specify which changes require reassessment; the passage of time or arrival of another prediction candidate does not itself establish a reason for HOLD. This proposed release checkpoint is distinct from ER validity and has not been validated by the laboratory.

Candidate sufficiency and execution integrity also matter on a collaborative path, where an authorized human works with AI assistance to assess the candidate and determine the appropriate action. Withholding automated authority does not itself authorize that alternative or remove the need to examine evidence and control execution. This RPVDN instantiation develops the automated Decision Node path; a collaborative route would require its own defined responsibilities and permissions.

Reassess Continuing Authority

Operating evidence feeds back into the justification for continuing authority. Candidate admission patterns, execution failures, and realized outcomes may expose weaknesses at different levels. Some observations can invoke prospectively specified controls; others require interpretive judgment. The same observation may warrant both responses.

Reassessing Standing. A changed eligible evidence state provides a prospective reason to reassess qualification. For RPVDN, LOOCV is proposed as the central empirical tool for characterizing the state on which the next prediction will depend. A materially changed state should be reassessed before inheriting autonomous qualification. Characterization can be reused while the relevant state and tested implementation remain unchanged.

The laboratory did not characterize every state encountered during the operational experiments. It directly examined State 0 and one transition to State 1. That transition illustrates the concern: under

Standard B, State 1 would fail qualification even though State 0 passed. Applied as an operating requirement, this reassessment could withhold authority before waiting for a subsequent prediction error. It does not establish that LOOCV will anticipate every deterioration in performance.

Realized sale price Y supplies a distinct, later feedback path by allowing a frozen prediction to be compared with its outcome. An illustrative rule withdrawing automated Execution Authority after two successive operational absolute percentage errors exceeding 5% shows how a specified trigger could act deterministically. Neither the threshold nor the sequence length is deployment-validated. A consequential observation may also warrant IS reconsideration before a numerical trigger is reached.

Candidate disposition and system authority must therefore remain distinguishable without becoming disconnected. A candidate-specific deficiency can require HOLD, correction, replacement, or escalation without revoking Standing for other eligible candidates. A recurring pattern or failure of a governing system requirement may instead require revocation of Standing and withdrawal of automated Execution Authority. Automated execution can resume only when the applicable qualification and authorization requirements are again satisfied. A candidate's acceptable result does not override a system-level loss of authority, and a candidate's failure does not by itself establish such a loss.

Continue Decision Node Engineering

The outer cycle asks what experience should change about the Decision Node itself. Accumulated evidence can reveal shortcomings in the model of reality, evidence representation, configured Decision Function, experimental methods, QC procedures, or governing standards. QC findings contribute to that broader assessment alongside other operating evidence. Excluded ERs, unsuccessful candidates, and unexpected outcomes can contribute to this work alongside qualifying results.

Continuing engineering turns those observations into questions that can be investigated. Does the evidence omit a consequential property characteristic? Does a representation encourage systematic misreading? Is the qualified domain too broad? Are the empirical tests sensitive to the changes that matter? Do admission and release controls detect the failures they are intended to address? Review methods must also address shared blind spots between production and QC. These questions can motivate revised configurations, modified QC procedures, further experiments, and better-supported standards.

Material changes require renewed qualification and authorization before automated execution under the revised configuration. Improved observed performance is evidence to assess; it is not self-authorizing permission. Preserving the lineage from evidence and specification through execution, QC, and disposition makes that reassessment open to inspection.

QC is foundational across both progressions: defects in evidence preparation, execution, or result handling can undermine the case for authority as well as an individual result. The DF Engineering, LOOCV, and QC companions develop those dependencies. The architecture assigns responsibility for responding to the evidence; it does not make persuasive outputs or successful calculations self-authorizing.

Scope and Limitations

The laboratory examined empirical performance, execution integrity, State 0, and one transition to State 1. It did not demonstrate the full automated governance progression shown here or characterize every state encountered in the prediction experiments. The map distinguishes proposed operating responsibilities from that demonstrated coverage.

The controls, retry and release rules, escalation responsibilities, and monitoring triggers require deployment-specific engineering and assessment. The illustrative two-successive-errors rule and Standards A and B are not deployment-validated thresholds. Human-led experimental checks and collaborative workbook reconciliation are not established autonomous QC capabilities.

The upper progression separates sufficiency judgments contributing to Standing; it is not a claim that evidence or interpretation always develops in a single linear pass. A candidate HOLD and revocation of system Standing are distinct responses. Neither candidate success nor affirmative human permission can override failure of governing engineering requirements.

This architecture is a starting point for further engineering and research. Its contribution is to make the responsibilities and their relationships explicit: Standing supports bounded authority, individual gates govern its use, and experience informs renewed justification and continuing Decision Node development.